



**College of Computing and Informatics
Department of Computer Science**

Course Syllabus

Course Title

GIS Programming Fundamentals

Course Code
1501532

Fall 2025/2026

Course Syllabus

Course Title	GIS Programming Fundamentals				
Course Code	1501532				
Credit Hours	3	Contact Hours	3		
Course Category	Program required				
Course Type	Compulsory				
Pre-requisite(s)					
Co-requisite(s)					
Semester	Fall 2025/2026				
Instructor Name	Email	Classroom	Section	Session Type	Time
Isam Mashhour Hasan Al Jawarneh	ijawarneh@sharjah.ac.ae	M1: Shari'ah & Law (Men)-125	61	Lecture Graduate	T 1700 - 1945

Course Description:

This course explains principles, syntax, and language elements associated with creating and running computer programming scripts for create GIS applications. Python scripting is used to efficiently run ArcGIS tools, read text files of data and ArcGIS attribute tables, applying vector data algorithms for GIS visualization, interact with map elements, manipulate batches of GIS data, and create basic user interfaces. Raster, network and surface data algorithms for GIS will also be introduced. Students completing this course should be able to implement a simple GIS workflow and build generalized application

Course Objectives/Goals:

The goals of this Course are to:

- Understand the fundamental concepts of programming and data visualization using Python
- Learn to use ArcGIS form simple read/write and visualization of the GIS data.
- Apply raster, surface, and network data algorithms on the GIS data

Course Learning Outcomes:

By the end of successful completion of this Course, the student will be able to:

1. Apply concepts of data structures and object-oriented programming for developing a GIS application
2. Use file input/output, vector data algorithm, and data visualization on GIS data
3. Illustrate ArcGIS tools and algorithms to be used with the GIS data
4. Apply raster, surface, and network data algorithms on GIS data.
5. Work in a team on a small GIS project

Alignment of Course Student Learning Outcomes to Program Student Learning Outcomes

Program Learning Outcomes (PLOs)	Course Learning Outcomes (CLOs)
Explain the scientific and theoretical concepts of geospatial data analysis and modeling.	- Use file input/output, vector data algorithm, and data visualization on GIS data - Illustrate ArcGIS tools and algorithms to be used with the GIS data

Program Learning Outcomes (PLOs)	Course Learning Outcomes (CLOs)
	- Apply raster, surface, and network data algorithms on GIS data.
Acquire, process and analyze geospatial data and appropriately display results.	- Use file input/output, vector data algorithm, and data visualization on GIS data - Illustrate ArcGIS tools and algorithms to be used with the GIS data - Apply raster, surface, and network data algorithms on GIS data.
Apply GIS and RS knowledge and algorithms in solving environmental, urban, social and economic problems.	- Apply concepts of data structures and object-oriented programming for developing a GIS application - Use file input/output, vector data algorithm, and data visualization on GIS data - Illustrate ArcGIS tools and algorithms to be used with the GIS data - Apply raster, surface, and network data algorithms on GIS data. - Work in a team on a small GIS project
Communicate both research and analysis results orally and in writing.	- Work in a team on a small GIS project

Weekly Distribution of Course Topics/Contents

Week	Topic	Comments*	CLOs
1	Introduction		1
2	Introduction to Python		1
3	Object Oriented Programming with Python		1
4	Language Control and File Input/output		2
5	Vector Data Algorithms		2
6	Vector Data Visualization		2
7	Shape File Handling		2
8	ArcGIS Programming		3
9	ArcGIS Programming		3
10	Raster Data Algorithms		4
11	Raster Data Algorithms		4
12	Network Data Algorithms		4
13	Network Data Algorithms		4
14	Surface Data Algorithms		4
15	Surface Data Algorithms		4
16	Project Submission and Final Exam		

Information on out-of-class assignments with due dates for submission

Assignment/Activity	Due Date	Comments
class Activity #1	Week 4	GIS Programming Basic
class Activity #2	Week 8	Vector Data Visualization

Assignment/Activity	Due Date	Comments
class Activity #3	Week 12	ArcGIS
class Activity #4	Week 15	Raster, Network, and Surface Data Algorithm

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Class Activity	Week 4, 8, 12, and 15	20
Midterm Exam	Week 8	20
Projects	Week 16	20
Final Exam	Week 16	40
Total		100

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS competencies	Other competencies
Apply concepts of data structures and object-oriented programming for developing a GIS application	- Case studies - Group Discussion - Lectures - Presentations	- Exams - Class Activity - Project Presentation - Project Report	Students should achieve a minimum of 75%	- Oral communication - Written communication - Intellectual curiosity and lifelong learning	
Use file input/output, vector data algorithm, and data visualization on GIS data	- Case studies - Lectures - Presentations	- Exams - Class Activity - Project Presentation - Project Report	Students should achieve a minimum of 75%	- Oral communication - Written communication - Intellectual curiosity and lifelong learning	
Illustrate ArcGIS tools and algorithms to be used with the GIS data	- Case studies - Lectures - Presentations	- Exams - Class Activity - Project Presentation - Project Report	Students should achieve a minimum of 75%	- Written communication - Intellectual curiosity and lifelong learning - Problem solving	

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS competencies	Other competencies
Apply raster, surface, and network data algorithms on GIS data.	- Case studies - Exercises - Lectures - Presentations	- Exams - Class Activity - Project Presentation - Project Report	Students should achieve a minimum of 75%	- Written communication - Teamwork or collaboration - Intellectual curiosity and lifelong learning - Problem solving	
Work in a team on a small GIS project	- Case studies - Exercises - Lectures - Presentations	- Exams - Class Activity - Project Presentation - Project Report - Research Paper	Students should achieve a minimum of 75%	- Teamwork or collaboration	

Scope and Nature of Assessment Tools:

The aim of this section is to clarify what is required and how to submit, grade and assess each assessment tool:

Exams:

This assessment tool is composed of two exams, a midterm exam (20%), and a final exam (40%). The midterm exam covers the first three learning outcomes of the course and covers the materials of the first five chapters. The final exam covers all course learning outcomes and all course materials with all chapters included.

Assignment:

The assignments assessment component is composed of four activity assignments which cover all the CLOs. This assessment tool aims at enabling the students to practice practical web design topics covered within the course materials.

Project:

The term project is an indispensable tool for assessing student's ability to work in a team project throughout the semester, covering all the practical topics discussed in the weekly schedule. It measures the student's ability to collaborate and be an active learner through an enhanced project-based learning experience.

Students are expected to write a high-quality research paper in one of the topics selected by the instructor. The project is divided into three parts. The combination of the three parts should result in a high-quality research paper of four to six pages at least.

Part 1: a 2-3 pages survey paper on the selected project topic. (Grade: 5%)

- Abstract
- Introduction
- Survey
- Future directions
- Conclusion

Part 2: Implementation, testing, and analysis of one of the techniques in the survey paper. (Grade: 10%)

- You need to do a project that is similar to one of the following scenarios, with new datasets of your selection.
- <https://pro.arcgis.com/en/pro-app/latest/help/analysis/networks/network-analyst-tutorials.htm>
- <https://desktop.arcgis.com/en/arcmap/latest/extensions/network-analyst/about-the-network-analyst-tutorial-exercises.htm>

Part 3: Do any of the following tasks. (Grade: 5%)

- Test your code on real data and compare with experimental results of the paper.
- Team up with a classmate and merge the two algorithms and make a hybrid-algorithm (combining both algorithms in such a way to enhance performance). Both students must have finished Part 2.
- Suggest an improvement to the algorithm, and then upon approval from instructor implement and test the improved code.

Each group then need to merge all the work into one original research paper and submit it to the instructor for approval and evaluation. Students are expected to do original work and produce novel results, comparing their work to at least one similar work from the related state-of-art and applying their algorithms (or solutions) to at least two related data sets!

Teaching and Learning Resources:

Textbook(s):

Yang, C. (2017). Introduction to GIS Programming and Fundamentals with Python and ArcGIS. 1st ed. CRC Press. [Link](#)

Recommended Readings:

Zandbergen, P. (2015). Python Scripting for ArcGIS. - ed. -.

Law, Mc. (2015). Getting to Know ArcGIS. 4th ed. American Society of Civil Engineers.

Other Resources:

Computer usage: Specialized software packages will be utilized during this course.

Attendance policy:

Attendance is compulsory. A student missing 25% of the total allocated course hours will be forced to withdraw in accordance with the university regulations.

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Notes:

There will be at least 4 in-class activities so that students can have a good practice of the language and the tools. A small team project will be completed at the end.

Grading rubrics:

Grading Rubric for Course Projects					
Scaled Score:		1/0	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
Introduction	10%	No clear introduction, absent, or no evidence.	The introduction states the main topic but does not adequately preview the structure of the report.	The introduction states the main topic and previews the structure of the report.	The introduction is engaging, states the main topic and previews the structure of the report.
Data	20%	Data are inadequate and selected following the wrong procedure.	Data are selected but procedure of data selection is not followed.	Data selected following the standard procedure but not described in report.	Data are correctly selected following standard procedure and adequately described in report.
Analysis and Reporting of Results (if applicable)	20%	Calculated wrong measures and analysis not relevant.	Calculated correct measures but analyses are not correct.	Calculated correct measures but analyses are not adequate.	Calculated correct measures and analyses are correct and provide more insights.
Authenticity and Citation*	20-30%	Some contents are paraphrased; plagiarism is most likely.	Some content is paraphrased; plagiarism is likely.	Some content is paraphrased and no suspicion of plagiarism.	All contents are original and well written.
Writing	20-30%	Errors in language that cause the content to be mostly incomprehensible.	Some errors in language that often inhibit comprehension.	Some errors in language that do not often inhibit comprehension.	Almost no error in language and well written.
Teamwork	10%	Team did not collaborate or communicate well.	Team did not collaborate or communicate well. A lack of mutual respect was frequently noted.	Team did collaborate or communicate well.	Team did collaborate or communicate well with high level mutual respect.

Grading Rubric for Course Assignments					
Scaled Score:		0/1	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
Focus: Purpose	10%	No awareness	Shows limited awareness of purpose.	Shows awareness of purpose.	Purpose is clear.
Main idea	10%	No main idea	Vague sense of a main idea, weakly supported throughout the paper.	There is a main idea supported throughout most of the paper.	Clearly presents a main idea and supports it throughout the paper.
Organization: Overall	5%	No sense of organization	There is a sense of organization, although some of the organizational tools are used weakly or missing.	Good overall organization, includes the main organizational tools.	Well-planned and well-thought out. Includes title, introduction, statement of main idea, transitions,

Grading Rubric for Course Assignments					
Scaled Score:		0/1	2	3	4
					and conclusion.
Organization: Paragraphs	5%	Paragraphs lack clear ideas	Some paragraphs have clear ideas, support from examples may be missing and transitions are weak.	Most paragraphs have clear ideas, are supported with some examples and have transitions.	All paragraphs have clear ideas, are supported with examples and have smooth transitions.
Content	50-60%	Content is not sound	Content is sound and solid; ideas are present but not particularly developed or supported; some evidence, but usually of a generalized nature.	Well-presented and argued; ideas are detailed, developed and supported with evidence and details, mostly specific.	Exceptionally well-presented and argued; ideas are detailed, well-developed, supported with specific evidence and facts, as well as examples and specific details.
Research (if assignment includes a research component)	10%	The paper does not use adequate research or if it does, the sources are not integrated well. They are not cited correctly according to MLA style, nor listed correctly on the Works Cited page.	Sources support some claims made in the paper, but might not be integrated well within the paper's argument. There may be a few errors in MLA style.	Sources are well integrated and support the paper's claims. There may be occasional errors, but the sources and Works Cited conform to MLA style sheet.	Sources are exceptionally well-integrated and they support claims argued in the paper very effectively. Quotations and Works Cited conform to MLA style sheet.
Style: Sentence structure, word, choice, tone	5%	Sentences aren't clear	Sentences are generally clear word usage generally appropriate, details are somewhat adequate but may have awkward structure or unclear content; there may be patterns of punctuation errors.	Sentences are clear, word usage appropriate, details are enough but may lack variation; a few may be awkward and there may be a few punctuation errors.	Sentences are clear, word usage appropriate; details are enough and varied in pattern, from simple to complex, with excellent use of punctuation.
Grammar and Mechanics	5%	Continuous errors	Shows a pattern of errors in spelling, grammar, syntax and/or punctuation. Could also be a sign of lack of proof-reading.	A few errors in grammar, spelling, syntax and punctuation, but not many.	Excellent grammar, spelling, syntax and punctuation.

Adapted from: <http://www.aur.edu/wp-content/uploads/2010/12/University-Writing-Rubric>

Grading Rubric for Case Studies					
Scaled Score:		0/1	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
<i>Clear explanation of key issues</i> - The problems, scope, and seriousness was clearly identified in the discussions of cases. - There was a well-focused diagnosis of key issues and problems that demonstrated a good grasp of the present situation - Effective concise executive summary	15%	Shows little understanding of the issues, key problems, and the entity's present situation and critical issues. Executive summary missing or poorly constructed.	Shows some understanding of the issues, key problems, and the entity's present situation and critical issues. Executive summary inadequate.	Shows adequate knowledge of the issues, key problems, and the entity's present situation and critical issues. Executive summary adequate.	Shows superior knowledge of the issues, key problems, and the entity's present situation and critical issues. Effective Executive Summary.
<i>Valid arguments; identification of relevant facts with supportive detail</i> - Logically organized, key points, key arguments, and important criteria for evaluating key facts were easily identified - Critical issues and key problems that supported the Case Analysis were identified and clearly analyzed and supported.	20%	Critical issues and key problems that supported the Case Analysis were poorly identified, analyzed, and supported.	Critical issues and key problems that supported the Case Analysis were not clearly identified, analyzed, and supported.	Critical issues and key problems that supported the Case Analysis were partially identified, analyzed, and supported.	Critical issues and key problems that supported the Case Analysis were clearly identified, analyzed, and supported.
<i>Appropriate analysis, evaluation, synthesis for the specific entity's environment identified</i> There was complete data on which to base a	40%	Analysis of key change drivers and the underlying the issues inadequate.	Analysis of key change drivers and the underlying the issues were not identified.	Analysis of key change drivers and the underlying the issues were partially identified.	Analysis of key change drivers and the underlying the issues were clearly identified.

Grading Rubric for Case Studies

Scaled Score:		0/1	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
thorough analysis • Key change drivers underlying the issues were identified. • Synthesis, analysis, and evaluations were clearly presented and supported in a literate and effective manner.					
Conclusions and recommendations are congruent with case analysis • Specific recommendations and/or plans of action provided. • Specific data or facts were referred to when necessary to support the analysis and conclusions. • Recommendations and conclusions were presented and supported in a literate and effective manner.	25%	Effective recommendations and/or plans of action not provided. Specific data or facts necessary to support the analysis and conclusions was not provided.	Effective recommendations and/or plans of action inadequate. Specific data or facts were not referred when necessary to support the analysis and conclusions.	Effective recommendations and/or plans of action were partially provided. Specific data or facts were occasionally referred when necessary to support the analysis and conclusions.	Effective recommendation, solutions, and/or plans of action were provided. Specific data or facts were referred when necessary to support the analysis and conclusions.

Grading Rubric for Course Projects

Scaled Score:		1/0	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
Introduction	10%	No clear introduction,	The introduction states the main topic	The introduction states the main	The introduction is engaging, states the

Grading Rubric for <i>Course Projects</i>					
Scaled Score:		1/0	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
		absent, or no evidence.	but does not adequately preview the structure of the report.	topic and previews the structure of the report.	main topic and previews the structure of the report.
Data	20%	Data are inadequate and selected following wrong procedure.	Data are selected but procedure of data selection is not followed.	Data selected following standard procedure but not described in report.	Data are correctly selected following standard procedure and adequately described in report.
Analysis and Reporting of Results (if applicable)	20%	Calculated wrong measures and analysis not relevant.	Calculated correct measures but analyses are not correct.	Calculated correct measures but analyses are not adequate.	Calculated correct measures and analyses are correct and provide more insights.
Authenticity and Citation*	20-30%	Some contents are paraphrased; plagiarism is most likely.	Some content is paraphrased; plagiarism is likely.	Some content is paraphrased and no suspicion of plagiarism.	All contents are original and well written.
Writing	20-30%	Errors in language that cause the content to be mostly incomprehensible.	Some errors in language that often inhibit comprehension.	Some errors in language that do not often inhibit comprehension.	Almost no error in language and well written.
Team work	10%	Team did not collaborate or communicate well.	Team did not collaborate or communicate well. A lack of mutual respect was frequently noted.	Team did collaborate or communicate well.	Team did collaborate or communicate well with high level mutual respect.

* If a case of plagiarism is established, the Course Project will be given zero grade and other actions will be taken according to the university bylaws.

Grading Rubric for Course Assignments					
Scaled Score:		0/1	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
Focus: Purpose	10%	No awareness	Shows limited awareness of purpose.	Shows awareness of purpose.	Purpose is clear.
Main idea	10%	No main idea	Vague sense of a	There is a main	Clearly presents a main

Grading Rubric for Course Assignments					
Scaled Score:		0/1	2	3	4
			main idea, weakly supported throughout the paper.	idea supported throughout most of the paper.	idea and supports it throughout the paper.
Organization: Overall	5%	No sense of organization	There is a sense of organization, although some of the organizational tools are used weakly or missing.	Good overall organization, includes the main organizational tools.	Well-planned and well-thought out. Includes title, introduction, statement of main idea, transitions, and conclusion.
Organization: Paragraphs	5%	Paragraphs lack clear ideas	Some paragraphs have clear ideas, support from examples may be missing and transitions are weak.	Most paragraphs have clear ideas, are supported with some examples and have transitions.	All paragraphs have clear ideas, are supported with examples and have smooth transitions.
Content	50-60%	Content is not sound	Content is sound and solid; ideas are present but not particularly developed or supported; some evidence, but usually of a generalized nature.	Well-presented and argued; ideas are detailed, developed and supported with evidence and details, mostly specific.	Exceptionally well-presented and argued; ideas are detailed, well-developed, supported with specific evidence and facts, as well as examples and specific details.
Research (if assignment includes a research component)	10%	The paper does not use adequate research or if it does, the sources are not integrated well. They are not cited correctly according to MLA style, nor listed correctly on the Works Cited page.	Sources support some claims made in the paper, but might not be integrated well within the paper's argument. There may be a few errors in MLA style.	Sources are well integrated and support the paper's claims. There may be occasional errors, but the sources and Works Cited conform to MLA style sheet.	Sources are exceptionally well-integrated and they support claims argued in the paper very effectively. Quotations and Works Cited conform to MLA style sheet.
Style: Sentence structure, word choice, tone	5%	Sentences aren't clear	Sentences are generally clear word usage generally appropriate, details are somewhat adequate but may have awkward structure or unclear	Sentences are clear, word usage appropriate, details are enough but may lack variation; a few may be awkward and there may be	Sentences are clear, word usage appropriate; details are enough and varied in pattern, from simple to complex, with excellent use of punctuation.

Grading Rubric for Course Assignments					
Scaled Score:		0/1	2	3	4
			content; there may be patterns of punctuation errors.	a few punctuation errors.	
Grammar and Mechanics	5%	Continuous errors	Shows a pattern of errors in spelling, grammar, syntax and/or punctuation. Could also be a sign of lack of proof-reading.	A few errors in grammar, spelling, syntax and punctuation, but not many.	Excellent grammar, spelling, syntax and punctuation.

Grading Rubric for Course Presentation				
Item	Exemplary	Accomplished	Developing	Beginning
Content	7	5	3	1
Introduction	Introduces the topic with clear objectives; employs a scenario which embeds useful information with a wide variety of data	Starts with clear objectives; employs the case study approach with useful information; may miss the most subtle data of the case	Uses an outline; presents a scenario having most obvious data; misses important data and information	Uses an outline but not accompanied with objectives; misses the inclusion of a scenario
Body of knowledge	Includes most relevant and significant information to explain the topic; elaborates on the pathologic changes accompanying the health problem; relates information to data in the scenario; demonstrates logical progression of information	Generally focuses on most relevant information but also attends to less pertinent ones; includes pathophysiology concepts; uses the case study in attempt to explain the scientific basis of data but misses the details of some important concepts; progresses in the presentation but with minor disturbance in flow	Makes an effort to focus on relevant information; uses pathophysiology concepts but has difficulty explaining the most subtle changes; does not link all changes to data in the scenario; at times information is not properly sequenced	Attempts to include relevant information but has difficulty explaining the pathologic changes of the health problem; appears not to know which data or information are (is) of utmost importance; fails to demonstrate logical progression of information
Presentation Skills	3	2	1	½
	Demonstrates	Generally displays confidence and	Is tentative in	Is hesitant or

Grading Rubric for Course Presentation				
	confidence and maintains good eye contact	maintains eye contact	displaying confidence and maintaining eye contact	ineffective when addressing the audience
	Ensures that audience can hear and see clearly	In most of the presentation, audience can hear and see clearly	Makes an effort to make the audience hear and see clearly	Has difficulty in making the audience hear and see clearly
	Uses appropriate and relevant presentation strategies to explore the topic	Uses appropriate strategies in most aspects of the presentation	Some of the used strategies lack educational value	Employs strategies that lack congruence with the presentation topic
	Does not read the entire content	Seeks to avoid reading the content	Reads items in the presentation	Reads almost all components of the presentation
	Finishes topic within allotted time	Is nearly close to finishing the topic on time	Needs 5 more minutes to finish the topics	Needs 10 more minutes to finish the topics
	Maintains adequate interaction with audience	Interacts with audience when clarifying some aspects of the presentation	Occasionally interacts with audience	Fails to interact effectively with audience
Total /32				

Same form is applied for self-assessment and peer assessment

Facilitator's Signature: ----- Student's Signature: -----